Efficient Multi-Session LiDAR SLAM with Sparse Marginalization and Adaptive Mapping

Benjamin A. Boyack and Joshua G. Mangelson

***Abstract*— We present an efficient multi-session LiDAR SLAM framework that adaptively switches between localization and mapping modes depending on whether the prior map provides sufficient constraints for the current session. We evaluate several mode switching policies spanning geometric, registration-based, graph-theoretic, and probabilistic frameworks, and propose a boundary-node covariance criterion for ensuring well-constrained graph regions across mode transitions. We introduce a sparse approximation to the marginalized factor that enables joint optimization of a fixed-lag smoother and a prior factor graph without the computational penalties of dense marginalization, while preserving the most spatially relevant prior constraints. Our experiments demonstrate that the proposed approach substantially reduces memory and computation relative to full SLAM while maintaining competitive localization and mapping accuracy.**

I. Introduction

As robotics advances toward large-scale, long-duration operations, efficient multi-session LiDAR SLAM has become increasingly important. In multi-session LiDAR SLAM, an initial map is typically constructed using traditional single-session methods [1]–[3]. Subsequent sessions then build upon this map by using inter-session loop closures to localize within the prior map and expand it as needed [4]–[6]. These algorithms are commonly formulated as estimation problems using factor graphs, which allows new session data to correct errors accumulated in previous sessions. However, naively fusing all measurements obtained into a single monolithic factor graph across all sessions leads to unbounded growth in computation time and memory usage. Furthermore, much of the information retained during localization is redundant when the prior map already provides sufficient constraints.

To address graph growth and computational cost, some methods [4], [7], [8] employ a fixed-lag smoother during localization in subsequent sessions, retaining only a recent window of states rather than the full history. In doing so, the question arises of whether or not to jointly optimize the fixed-lag-based localization with the prior map. If the fixed-lag graph is jointly optimized with the prior graph, marginalizing old nodes produces a dense information matrix, significantly increasing optimization time. Conversely, if the fixed-lag graph is optimized separately from the prior graph, errors in the prior map propagate into the current session and cannot be corrected.

In this paper, inspired by [9], we address both of these issues. In the first session, a map is built using standard LiDAR SLAM techniques. In subsequent sessions, our algorithm adaptively switches between one of two modes: *localization* or *mapping*, as illustrated in Fig. 1. In localization mode (blue), a fixed-lag smoother is used while the prior map provides sufficient coverage. When the prior map becomes insufficient, due to environmental changes or entry into a previously unmapped area, the system transitions to mapping mode (green), where all new nodes are retained and incorporated into the prior map for future sessions. The transition between modes (red) is governed by one of several *switching policies*. Unlike [9], we jointly optimize the prior map and the current session in a single factor graph across both modes. To prevent the density introduced by marginalization from degrading performance, we apply a sparsification method to the resulting marginalized factors.

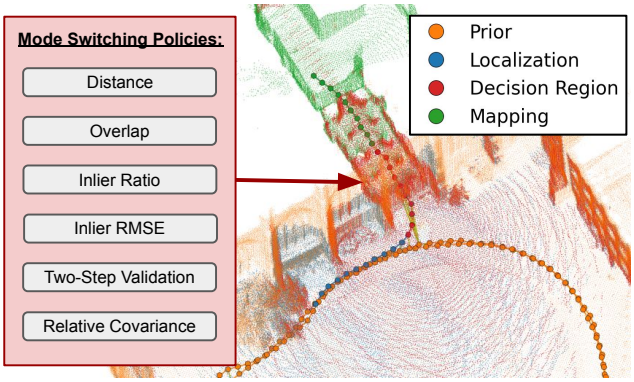


Fig. 1. A prior map (orange) is constructed using traditional LiDAR SLAM. During a subsequent session, the system localizes within the prior map using a fixed-lag smoother (blue), retaining only a recent window of poses. When the prior map becomes insufficient for reliable localization, the system transitions to mapping mode (green) at a mode-switching boundary (red), retaining all new information for future sessions. The switching boundary is determined by one of the policies evaluated in this work, each offering different trade-offs between computational cost, memory usage, and accuracy.

Our contributions are as follows:

- We present a factor-graph-based multi-session LiDAR SLAM framework that adaptively switches between fixed-lag localization and augmentation of a prior map. This is made possible via a novel sparsification method that maintains computational efficiency during joint optimization of a fixed-lag smoother and a prior map.
- We present an evaluation of several switching metrics and a novel relative covariance metric for detecting when the prior map provides insufficient localization support.
- We present a general framework for preserving strong graph connectivity across mode transitions, independent

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B. Boyack, J. Mangelson, are at Brigham Young University.
{bab272, joshua_mangelson}@byu.edu.
We open source our code at [TODO: Insert link to github repo].

of the choice of switching policy.

II. Related Work

A. Multi-Session and Lifelong Mapping

Multi-session mapping merges a prior map with new observations through inter-session loop closures and map expansion, while lifelong mapping extends this by additionally incorporating dynamic object detection, removal, and map refinement to continuously maintain a globally consistent map. LT-Mapper [10] established the canonical three-module lifelong framework: multi-session SLAM, dynamic object removal, and map updating. The multi-session component — the focus of this work — has since been refined independently of the lifelong pipeline: MS-Mapper [6] models point-level Gaussian distributions and incorporates new observations based on a Wasserstein distance criterion. LL-Localizer [5] prioritizes efficiency through an incremental octree structure. The authors of [11] replace submap regions of the prior map when scan-matching performance degrades.

B. Graph Sparsification in Long-Term SLAM

Graph-based methods dominate multi-session SLAM due to their ability to bridge large spatial and temporal gaps, but unbounded graph growth creates challenges for memory usage and real-time optimization. Marginalization is the standard tool for controlling this growth, but eliminating variables introduces fill-in that densifies the factors connecting the remaining nodes. GLC [12] approximates these marginalized constraints with sparse linear factors. NFR [13] approximates marginalized nodes/factors using a new nonlinear factor that is designed to minimize the Kullback–Leibler divergence between the dense marginalization and its sparse approximation. The work in [14] applies ℓ_1 regularization to sparsify edges in a consistent manner. From a spectral perspective, [15] shows that maximizing the algebraic connectivity of the measurement graph allows edges to be removed while bounding the resulting estimation error. The densification problem is particularly acute in fixed-lag smoothing over a prior graph, where joint optimization introduces a dense n-ary factor over all marginalized prior nodes, increasing optimization time and threatening real-time operation. We propose a sparse approximation to this marginalized factor that maintains the benefits of joint optimization while preserving computational efficiency.

C. Adaptive Transition Between Localization and Mapping

Recent multi-session SLAM methods define explicit localization and mapping modes depending on whether the prior map provides sufficient constraints for the current session [4], [9]. During localization, a fixed-lag smoother limits computational cost by retaining only a recent window of states [7], [8]. Methods that optimize the fixed-lag graph independently of the prior graph treat the prior as immutable, which prevents errors in the prior from being corrected during the current session [8], [9]. Joint optimization [4] relaxes this assumption but at the cost of the graph densification discussed above. In this work, we jointly optimize both sessions across both modes and apply sparsification to the marginalized factors to control the computational cost introduced by densification.

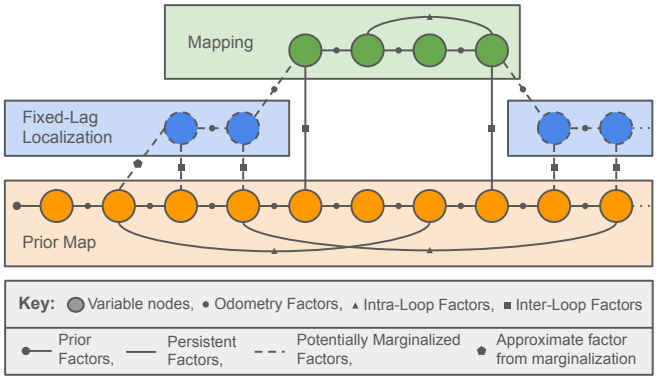


Fig. 2. Proposed jointly optimized factor graph. The prior map is constructed using traditional SLAM methods. Localization and mapping nodes are added during subsequent sessions, where localization nodes are marginalized out in a fixed-lag fashion while mapping nodes are retained in the persistent map.

D. Graph Connectivity at Mode Transitions

The boundary between localization and mapping modes represents the weakest point in the combined graph. As noted in [9], inter-session connections at this boundary are especially sparse, which can cause map warping between the prior and newly mapped regions. Placed and Castellanos [16] show that the graph Laplacian approximates the Fisher Information Matrix at reduced computational cost, enabling graph connectivity metrics to serve as proxies for estimation quality. Building on this, [9] uses the Laplacian spectrum to detect when the current session provides more information than the prior map and triggers the transition to mapping mode. We address the boundary connectivity problem by retaining additional nodes near the mode transition to maintain a well-connected graph across sessions. We additionally evaluate several switching metrics for triggering this transition and propose a novel relative covariance metric.

III. Efficient Localization and Adaptive Mapping

We model our multi-session optimization using a single factor graph conceptually divided into three sections: prior, localization, and mapping. Each section contains variable nodes V (representing the robot's SE(3) poses) and factor nodes F (representing sensor constraints). The prior nodes (V_p, F_p), together with their associated LiDAR scans, are inherited from previous sessions and constitute the prior map. Localization nodes (V_{loc}, F_{loc}) are acquired in the current session while operating in localization mode, and mapping nodes (V_{map}, F_{map}) are acquired while operating in mapping mode. Consecutive poses are constrained by odometry factors F_{odom}, while previously visited areas, whether in the prior map or a newly mapped region, are constrained by loop closure factors F_{lc}. We jointly optimize over the variable nodes associated with each of these sections in a single factor graph as illustrated in Fig. 2.

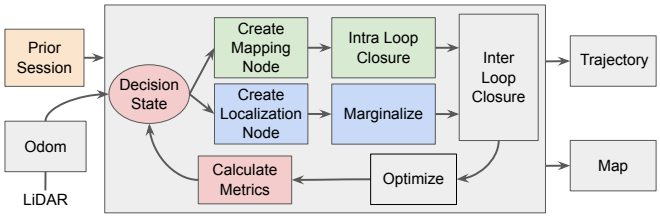


Fig. 3. System pipeline. Odometry between keyframes is obtained using KISS-ICP and provided to the factor graph, where each new node is added as either a localization or mapping node according to the current decision state. In localization mode, fixed-lag marginalization is performed. In mapping mode, mapping nodes are retained for incorporation into the prior map and intra-session loop closure detection is performed. Regardless of the decision state, inter-session loop closure detection is performed before graph optimization. Following optimization, the switching policy is evaluated to determine whether the decision state should remain unchanged or switch for the next timestep.

In localization mode, the most recent N nodes are retained in a fixed-lag smoother; nodes that fall outside the lag window are marginalized and replaced by a marginal factor f_{marg}. When localization is determined to be insufficiently reliable, the system transitions to mapping mode, where newly acquired nodes are retained without marginalization. The operating mode at time t is represented by

$$s^t \in \{\mathrm{Loc}, \mathrm{Map}\}. \tag{1}$$

The mode at the subsequent time step is determined by a switching policy according to

$$s_{t+1} = \pi(\mathrm{m}^t, \tau), \tag{2}$$

where m^t denotes one or more scalar metrics describing the current localization and mapping state and τ denotes the corresponding decision threshold(s). The policies evaluated in this work draw from geometric, registration-based, graph-theoretic, and probabilistic frameworks to determine when the system should transition between modes.

An overview of the system pipeline is shown in Fig. 3. We use KISS-ICP [3] as our LiDAR odometry front-end due to its speed and simplicity, though any LiDAR or LiDAR inertial odometry method can be substituted. At each time step, new odometry factors are added to F_{loc} or F_{map} depending on s^t. Marginalization is then performed if any localization nodes fall outside the lag window. Once the lag window contains only mapping nodes, marginalization ceases and intra-session (same-session) loop closure detection is performed instead. This is followed by inter-session (between-session) loop closure detection, joint optimization of the full graph, and an update of the switching metric given the current estimated state.

IV. Joint Optimization of Fixed-Lag Smoother on Prior Factor Graphs

As described in Section III, inter-session loop closure factors between localization nodes V_{loc} and prior nodes V_p constrain the current session to the prior map during localization mode. As localization nodes leave the lag window, they are marginalized using the Schur complement, producing a marginal factor f_{marg} that connects the remaining variables in the marginalized node's Markov blanket. In general, f_{marg} may therefore connect the oldest retained localization node to multiple prior nodes through the inter-session loop closures accumulated during localization.

For a system repeatedly traversing regions represented in the prior map, successive marginalization operations cause the connectivity of f_{marg} to grow. In the limiting case, a trajectory that establishes loop closures throughout a previously mapped route can result in f_{marg} connecting to most nodes in V_p, producing an increasingly dense information matrix that raises the computational cost of each subsequent optimization. We therefore introduce an approximation that limits the number of variables connected by f_{marg} while retaining the most spatially relevant prior connections.

Let m denote the number of nodes currently connected by f_{marg}, and let M denote the maximum allowable number of connected nodes. We seek to maintain

$$m \leq M \tag{3}$$

where, for example, $M = 2$ restricts f_{marg} to a binary factor. When $m > M$, the dense marginal factor is replaced by an approximate factor connecting only M nodes. Of the m nodes in the Markov blanket of f_{marg}, we retain the last node in the lag window and the $M - 1$ prior nodes within the Markov blanket that are spatially nearest to it. This spatial criterion provides a deterministic and computationally inexpensive proxy for information relevance, preserving connections to the portion of the prior map most closely associated with the current trajectory.

To construct the approximation, let Λ^+ denote the information matrix of the original dense marginal factor. We partition the nodes connected by f_{marg} into a retained set, consisting of the M selected nodes, and a removed set, consisting of the remaining $m - M$ nodes whose direct connection to f_{marg} will be eliminated. Importantly, this partition does not remove those nodes from the graph; it only determines which nodes remain directly connected by the replacement factor. Partitioning Λ^+ accordingly gives

$$\Lambda^+ = \begin{bmatrix} \Lambda_{rr} & \Lambda_{rk} \\ \Lambda_{kr} & \Lambda_{kk} \end{bmatrix} \tag{4}$$

where Λ_{kk} contains the information associated with the M retained nodes, Λ_{rr} contains the information associated with the $m - M$ removed nodes, and Λ_{rk} and Λ_{kr} capture the cross-correlations between the two sets. The information from the removed nodes is folded into the retained set via the Schur complement:

$$\tilde{\Lambda} = \Lambda_{kk} - \Lambda_{kr}\Lambda_{rr}^{-1}\Lambda_{rk}. \tag{5}$$

The original dense f_{marg} is then replaced by a new factor connecting only the M retained nodes with information matrix $\tilde{\Lambda}$. All other nodes and their associated factors remain unchanged in the graph, so the approximation reduces factor connectivity without discarding any variable nodes.

V. Graph Switching Policies

In this section, we discuss several different methods that can be used for switching between localization and mapping

modes. We define the policy function in (2) as follows:

$$\pi_i(\mathrm{m}^t, \tau_i) = \begin{cases} \text{Loc} & \text{if } \delta_i \mathrm{m}^t < \delta_i \tau_i, \\ \text{Map} & \text{otherwise,} \end{cases}$$

where subscript i indexes the different policies, $\mathrm{m}^t = g_i(\hat{V}^t, F)$ is the current policy's metric value computed from estimated poses $\hat{V}^t$ and graph factors F, τ_i is the decision threshold, and $\delta_i = 1$ if a smaller m^t favors localization and $\delta_i = -1$ otherwise. Each policy is fully specified by defining g_i, τ_i, and δ_i.

We group the switching metrics into four categories: geometric, registration-based, graph-theoretic, and probabilistic.

A. Geometric

1) Distance: The distance policy computes the Euclidean distance between the current robot position x^t and the closest prior node, making it a simple and versatile baseline that requires only the current and prior estimated poses:

$$g_{dist} = \min_{x_j \in V_p} d(x^t, x_j), \quad \tau_{dist} \in [0, \infty), \quad \delta_{dist} = 1$$

where $d(\cdot, \cdot)$ denotes the Euclidean distance between two poses and τ_{dist} is the maximum distance threshold beyond which mapping mode is triggered.

2) Overlap: The overlap policy was introduced by [4] for lifelong localization using a grid-based decomposition of the prior map into submaps. It evaluates the spatial overlap between a region associated with the current pose and the nearest prior submap, and is well suited to grid-based representations where the spatial extent of each submap is predefined. However, it implicitly assumes that each grid cell is sufficiently populated to support reliable localization, and the formulation is inherently planar, making it unable to distinguish between spatially separated regions with similar horizontal extents, such as different floors of a building.

We generalize the overlap policy to arbitrary submap representations by deriving spatial extents directly from point cloud data. Let $b(\cdot)$ denote a bounding-box function that returns the 1st-to-99th percentile bounds of the associated LiDAR points, defining the corresponding rectangular region. Let L^t denote the point cloud associated with the current pose x^t, and let S_p denote the closest prior submap, constructed by aggregating the point clouds of the k prior nodes nearest to x^t. The intersection of the two bounding regions is then

$$B_\cap = b(L^t) \cap b(S_p).$$

The overlap measure is defined as the area of this intersection normalized by the area of the current region:

$$g_{overlap} = \frac{\mathrm{Area}(B_\cap)}{\mathrm{Area}(b(L^t))}$$
$$\tau_{overlap} \in [0, 1], \quad \delta_{overlap} = -1$$

Thus, $g_{overlap}$ represents the fraction of the current region that overlaps the prior submap. Using percentile-based bounds reduces the influence of outlying LiDAR returns while providing a data-driven estimate of each region's spatial extent.

B. Registration

Point cloud registration estimates the transformation between two point clouds by minimizing a correspondence-based objective. Following the convention used by Open3D [17], we denote the source and target point clouds as P and Q, respectively, with points $p_i \in P$ and $q_i \in Q$. In our application, P represents the current LiDAR scan and Q represents a local submap extracted from the prior map.

1) Inlier Ratio: The inlier ratio measures the fraction of points in the current LiDAR scan that form valid correspondences with the prior map. Let $\mathcal{I}$ denote the inlier correspondence set, $N_\mathcal{I}$ the number of inlier correspondences, and N_P the number of points in P. The inlier ratio is defined as:

$$g_{ratio} = \frac{N_\mathcal{I}}{N_P}, \quad \tau_{ratio} \in [0, 1], \quad \delta_{ratio} = -1$$

A larger inlier ratio indicates that a greater proportion of the current scan is geometrically consistent with the prior map, providing a direct measure of how well the current observation is supported by the prior. In practice, τ_{ratio} is selected based on the minimum fraction of the scan required to participate in valid correspondences, typically in the range $[0.5, 0.8]$.

2) Inlier RMSE: The inlier RMSE measures the geometric residual of the accepted correspondences, defined as:

$$e_{rmse} = \sqrt{\frac{1}{N_\mathcal{I}} \sum_{i \in \mathcal{I}} \|p_i - q_i\|^2}$$

A low e_{rmse} alone is not a sufficient indicator of strong registration. If $N_\mathcal{I}$ is small, the residual may be artificially low due to insufficient correspondence support. A registration is therefore only considered strong when e_{RMSE} is small and $N_\mathcal{I}$ is large. Let $\tau_\mathcal{I}$ denote the minimum required number of inlier correspondences. The metric is then defined as:

$$g_{rmse} = \begin{cases} e_{RMSE} & \text{if } N_\mathcal{I} > \tau_\mathcal{I} \\ \infty & \text{otherwise} \end{cases}$$
$$\tau_{rmse} \in [0, \infty), \quad \delta_{rmse} = 1$$

Lower values indicate greater geometric consistency among the accepted correspondences. The inlier ratio and inlier RMSE provide complementary measures of registration quality: a high inlier ratio indicates broad coverage of the current scan by the prior map, while a low inlier RMSE indicates accurate alignment of the accepted correspondences. In practice, τ_{ratio} is more intuitive and easier to tune, whereas τ_{rmse} is more sensitive to the specific environment.

C. Graph Theoretic

Following the notation of [9], the graph Laplacian $\mathbf{L} \in \mathbb{R}^{m \times m}$, where m is the number of nodes in the graph, encodes the connectivity between variable nodes through their connecting factors. The eigenvalues of $\mathbf{L}$ characterize graph connectivity: the smallest eigenvalue is always zero, and the second-smallest, λ_2, known as the algebraic connectivity, measures how well connected the graph is. A value of $\lambda_2 = 0$ indicates a disconnected graph.

A complementary measure of connectivity is the average node degree,

$$d \triangleq \frac{1}{m}\sum_{i=1}^{m} \ell_{ii}, \quad \ell \in \mathbf{L}$$

which reflects the density of connections within the graph. In multi-session mapping, an increase in loop closure constraints generally increases d, while a graph consisting primarily of odometric constraints yields a lower degree.

1) Two-Step Validation: The two-step validation metric was introduced by [9] to detect potential graph disconnections and weakly connected regions using d and λ_2. The method constructs Laplacian matrices with edges weighted by the corresponding factor information matrices and uses their trace and minimum eigenvalue to compute normalized measures $\bar{d}$ and $\bar{\lambda}_2$, respectively. The first stage compares $\bar{d}$ against its moving average; when $\bar{d}$ drops more than 2σ below the mean, the second stage evaluates whether $\bar{\lambda}_2 = 0$ or $\bar{d} < 4\sigma$, with either condition triggering mapping mode.

Reproducing this procedure exactly is challenging because the implementation is not publicly available and several heuristics are not fully specified. We were unable to consistently reproduce the condition in which $\bar{\lambda}_2 = 0$ while $\bar{d} < 2\sigma$, and consequently the two-step procedure did not trigger mapping mode consistently when the robot left the previously mapped region.

To evaluate the underlying graph-theoretic criteria independently of this discrepancy, we also consider a simplified variant (which we call *two-option*) that evaluates the second-stage conditions directly, without the first-stage gating step. Mapping mode is triggered when either $\bar{\lambda}_2 = 0$ or $\bar{d} < 4\sigma$:

$$g_{option} = \begin{cases} 1 & \text{if } \bar{\lambda}_2 = 0 \text{ or } \bar{d} < 4\sigma \\ 0 & \text{otherwise} \end{cases}$$
$$\tau_{option} = 0.5, \quad \delta_{option} = 1.$$

Removing the first-stage gating condition requires $\bar{\lambda}_2$ to be evaluated at every time step rather than only when the first stage is satisfied, which incurs additional computational cost relative to the original procedure.

D. Probabilistic

We investigate whether the relative uncertainty between the current pose estimate x^t and a prior node can provide a useful indicator of when localization uncertainty has grown sufficiently large to warrant transitioning to mapping mode.

1) Relative Covariance: Relative covariance in the Lie algebra has been characterized in [18] using a left-invariant convention. Since GTSAM [19] adopts a right-invariant convention, we follow the transformation presented in [20] to obtain the corresponding expression. The relative covariance Σ_{rel} between nodes i and j is given by

$$\Sigma_{rel} = \alpha\Sigma_{ii}\alpha^T + \Sigma_{jj} - \alpha\Sigma_{ij} - \Sigma_{ij}^T\alpha^T$$
$$\alpha = \mathrm{Ad}_{x_j^{-1}}\,\mathrm{Ad}_{x_i},$$

where Σ_{ii} and Σ_{jj} are the marginal covariances of poses x_i and x_j, Σ_{ij} is their cross-covariance, and Ad_x denotes the adjoint associated with the respective pose estimate. In our application, node i is the prior node closest to the current pose, and node j is the most recent node in the current session, so Σ_{rel} characterizes the uncertainty of the current pose relative to its nearest anchor in the prior map.

To obtain a scalar measure suitable for the switching decision, we use the log-determinant of the relative covariance:

$$g_{rel} = \log\det(\Sigma_{rel}), \quad \tau_{rel} \in [0, \infty), \quad \delta_{rel} = 1$$

The determinant captures the generalized volume of the uncertainty ellipsoid, and the logarithm compresses the dynamic range into a numerically convenient scalar. Larger values of g_{rel} therefore indicate greater relative uncertainty between the current estimate and the prior map, suggesting that the prior is no longer providing sufficient constraint to support reliable localization.

VI. Maintaining Well-Connected Graphs

The connection between prior nodes V_p and newly acquired mapping nodes V_{map} can become weak during transitions between localization and mapping modes. The metrics described in Section V do not guarantee loop closures between V_p and V_{map}, because loop closures between V_{loc} and V_p are marginalized as localization nodes leave the fixed-lag window. Consequently, the newly mapped region may become weakly constrained relative to the prior map, which can lead to deformation of the estimated trajectory and map. The method in [9] addresses graph connectivity through the average node degree $\bar{d}$; however, under their two-step validation procedure, a weakly connected graph is only detected under specific conditions involving both $\bar{d}$ and $\bar{\lambda}_2$. We instead monitor the marginal covariance of *boundary nodes*, the first and last mapping nodes associated with a newly explored region, as a proxy for whether the graph remains sufficiently constrained, providing a mechanism that can be applied independently of the choice of switching policy.

The top of Fig. 4 illustrates the evolution of the determinant of the marginal covariance of a boundary node at the start of a mapping transition. As localization nodes V_{loc} and their associated loop closure edges are marginalized, the uncertainty of the boundary node increases substantially. Once all localization nodes have been marginalized, the boundary-node covariance remains large due to the absence of direct constraints between V_{map} and V_p. In the limiting case where $M = 1$ in (3), the marginal factor becomes unary and provides no connection between the mapping and prior nodes, potentially resulting in a disconnected graph.

To prevent the graph from becoming weakly connected, we monitor the relative change in the determinant of the boundary-node marginal covariance, illustrated in the bottom of Fig. 4. We define this quantity as

$$\beta = \frac{\det(\Sigma_b^t) - \det(\Sigma_b^{t-1})}{\det(\Sigma_b^{t-1})}, \tag{6}$$

where Σ_b denotes the marginal covariance of the boundary node. The proposed mechanism reclassifies localization

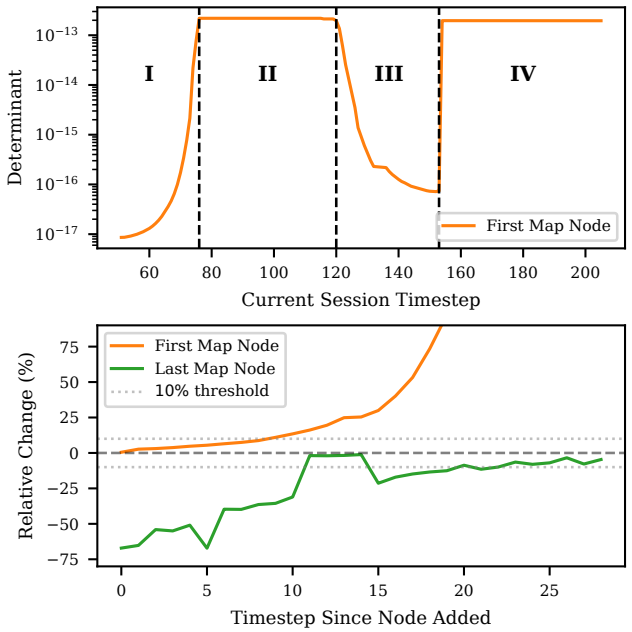


Fig. 4. Top: evolution of the determinant of the marginal covariance of a boundary node during a localization-to-mapping transition. (I) Localization nodes are marginalized as mapping proceeds. (II) All localization nodes have been marginalized while mapping continues. (III) The robot enters a previously mapped region and localization nodes with loop closures restore connectivity to the newly mapped nodes. (IV) These localization nodes are subsequently marginalized. Bottom: relative change in the determinant of the boundary-node covariance during the transition from localization to mapping (orange) and from mapping back to localization (green).

nodes as mapping nodes before the boundary becomes weakly constrained, assuming that the lag window is long enough that the transition to mapping mode occurs before all inter-session loop closures have been marginalized out of the window. Specifically, for the boundary at the start of a mapping region, if

$$\beta > 0.1, \tag{7}$$

all localization nodes preceding the boundary node are reclassified as mapping nodes, preserving additional constraints between the newly mapped region and the prior map. Similarly, for the boundary at the end of a mapping region, if

$$-0.1 < \beta < -0.01, \tag{8}$$

all localization nodes following the boundary node are reclassified as mapping nodes. Here, a negative value of β indicates a decreasing boundary-node covariance as the robot returns to a previously mapped region and additional loop closure constraints become available. A value of $\beta \approx 0$ at the end boundary does not necessarily indicate sufficient constraint: if newly acquired localization nodes do not establish loop closures with the newly mapped region, the covariance may remain approximately unchanged despite the absence of constraining information. The boundary-node criterion therefore uses a finite negative range for β rather than treating $\beta \approx 0$ as evidence of sufficient connectivity. The thresholds 0.1 and -0.1 were chosen experimentally to err on the side of retaining additional constraints before the covariance grows substantially, and may be tuned for other environments or lag window sizes.

VII. Evaluation

We evaluate our approach on the North Campus Long-Term (NCLT) dataset [21] and the Newer College Extension (NCE) dataset [22], which provide complementary evaluation scenarios: NCLT is used to evaluate computational resource usage due to its long duration and large spatial extent, while NCE is used to evaluate mapping accuracy due to its large initially-unmapped regions.

For all experiments, we use the SLAM pipeline shown in Fig. 3, varying only the switching metric, so that differences in performance can be attributed to the metric rather than to the underlying pipeline. We compare all switching metrics against a *full SLAM* baseline in which mapping mode is always active, corresponding to the conventional approach used by traditional multi-session SLAM methods. All thresholds were selected experimentally: $\tau_{dist} = 10$, $\tau_{overlap} = 0.7$ (following [4]), $\tau_{ratio} = 0.7$, $\tau_{rmse} = 0.24$ with $\tau_{\mathcal{I}} = 1000$, and $\tau_{rel} = -38$ (relating to a covariance matrix with standard deviations 0.1 m for translation and 0.017 rad for rotation). In practice, the graph-theoretic policies are threshold-free, as their switching conditions are defined by the σ-based criteria described in Section V.

We first evaluate the proposed marginal approximation using the log-determinant of the estimated covariance as a measure of uncertainty, then evaluate the switching policies in terms of execution time, memory consumption, and localization and mapping accuracy. All experiments are performed using the GTSAM [19] optimization library on a laptop equipped with an Intel Core i7-12850HX CPU and 32 GB of RAM.

A. Fixed-Lag Marginal Approximation

We evaluate the f_{marg} approximation proposed in Section IV using the NCE dataset, with the prior map constructed from the quad-easy sequence and the quad-hard sequence used as the current session. No switching policy is applied, keeping the system in localization mode throughout and isolating the behavior of the approximation.

We compare three configurations using a fixed-lag window of 10 nodes: 1) full SLAM without marginalization, providing the reference covariance; 2) fixed-lag smoothing with the exact dense marginal factor; and 3) fixed-lag smoothing with the proposed approximation. Figure 5 shows the log-determinant of the covariance of the most recent node over the quad-hard sequence, along with the per-timestep difference

$$\Delta_k = \log\det(\Sigma_{\text{approx},k}) - \log\det(\Sigma_{\text{dense},k}), \tag{9}$$

where $\Delta_k = 0$ indicates equal covariance determinants between the two methods. The proposed approximation produces a larger covariance determinant than the exact dense marginalization at 100% of timesteps after marginalization begins, indicating that the approximation is consistently conservative with respect to uncertainty volume rather than introducing overconfidence.

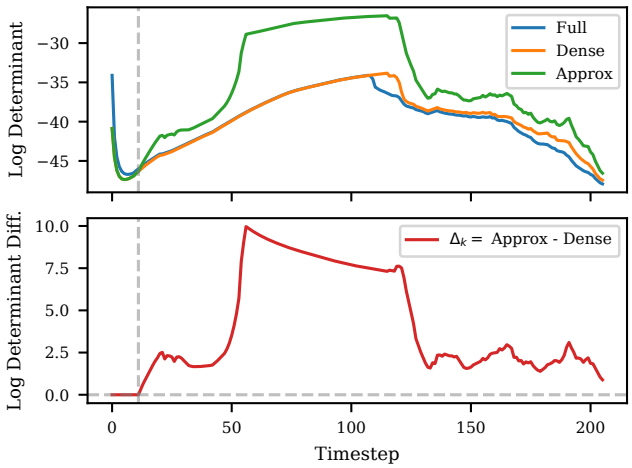


Fig. 5. Log-determinant of the covariance of the most recent graph node. Blue: full SLAM without marginalization (reference). Orange: fixed-lag smoothing with exact dense marginalization. Green: fixed-lag smoothing with the proposed sparse approximation. The vertical dotted line marks where the fixed-lag window fills and marginalization begins. The increase in green during iterations 50–125 corresponds to the robot exploring an unmapped region. Bottom: per-timestep difference Δ_k between the approximation and exact dense marginalization; the horizontal dotted line indicates $\Delta_k = 0$.

B. Policy Resources and Accuracy

1) Resources: We evaluate the computational resource requirements of each policy using the NCLT dataset, reporting mean and standard deviation of optimization time and metric computation time, as well as final graph size and memory for both the graph and map. Metric computation time is omitted for the inlier ratio and inlier RMSE policies since the metric is computed during inter-loop registration earlier in the pipeline; the Laplacian matrix is updated incrementally for the graph-theoretic policies. Results are shown in Table I.

The distance, inlier ratio, and inlier RMSE policies all have low metric computation costs, with the registration-based policies incurring negligible additional cost due to registration result caching. The relative covariance policy requires more computation than these but remains efficient due to GTSAM's incremental covariance retrieval, with cost scaling modestly with map size. The overlap policy requires more time due to our data-driven submap boundary implementation.

The two-step validation policy requires substantially more computation than the other policies, with an overall average requirement of approximately 0.8 s. However, most of this time is due to the computation of $\bar{\lambda}_2$ during mapping mode, which requires approximately 11.6 s. In contrast, during localization when $\bar{\lambda}_2$ is not calculated, only 82 ms are required on average.

All switching policies substantially reduce graph and map memory relative to full SLAM, reducing node count to approximately half of the original size with graph and map memory directly proportional to final node count.

2) Accuracy: We evaluate localization and mapping accuracy using the NCE dataset, processing sequences sequentially quad-medium (QM), quad-hard (QH), cloister, and park — each building upon the map produced by the preceding sequence, with the quad-easy (QE) sequence as the prior session. The quad-hard, cloister, and park sequences progressively introduce larger unmapped regions, providing increasingly challenging switching scenarios. We compare full SLAM against each switching policy under two boundary conditions: the exact policy-defined boundary and the well-constrained boundary proposed in Section VI. Accuracy is evaluated using absolute trajectory error (ATE) via the *evo* package [23], with results reported in Table II.

TABLE I

OPTIMIZATION TIME, METRIC COMPUTATION TIME, GRAPH SIZE, AND RAM MEMORY USING NCLT WITH 2012-01-08 AS THE PRIOR SESSION AND 2012-01-15 AS THE CURRENT SESSION.

Policy	Computation Time (ms)		Nodes	Graph (MB)	Map (GB)
	Opt.	Metric			
Full	313.6±173.4	-	12285	16.66	15.80
Distance	76.0±16.3	**1.2±0.7**	6225	6.29	7.94
Overlap	70.3±13.3	12.7±3.3	6082	6.06	7.77
In. Ratio	80.1±13.5	*	**5995**	**5.83**	**7.65**
In. RMSE	79.9±13.7	*	6013	5.85	7.67
Two Step	141.4±72.3	859±2899	9690	12.32	12.30
Rel. Cov.	**60.0±23.3**	4.0±1.6	6596	7.34	8.39

Bold: best. Underlined: second best.
*Metric computed during registration; timing omitted.

During quad-medium, only the two-step validation and two-option policies transition to mapping; all other policies remain in localization mode for the duration of the sequence and thus incur no additional error. For the graph-theoretic policies, the 4σ criterion can be triggered by small deviations between the current session and the prior session even when the prior session provides adequate localization, resulting in unnecessary mapping during quad-medium.

As sequences progressively explore unmapped regions, the choice of boundary condition becomes increasingly important. With exact policy-defined boundaries, several policies accumulate substantially more error than full SLAM, most notably the registration-based policies (see Table II). The well-constrained boundary reduces ATE for every evaluated policy, with the most pronounced improvements for inlier ratio and inlier RMSE, due to insufficiently constrained localization-to-mapping transitions with the exact boundary.

Overall, the switching policies substantially reduce the amount of mapping required while maintaining competitive localization accuracy, provided the transition boundary is sufficiently constrained. The well-constrained boundary is particularly important when the system leaves the prior map, where weak connections between the prior and newly mapped graph regions can otherwise lead to significant accumulated error.

VIII. CONCLUSION

We developed an efficient multi-session LiDAR SLAM system that uses a fixed-lag smoother during localization and approximates the resulting marginalized factor to maintain a sparse graph, avoiding the redundant storage of information already represented in the prior session. We evaluated several switching policies for determining when the prior session becomes insufficient and the system should transition between

TABLE II

ATE FOR NEWER COLLEGE DATASET COMPARED AGAINST FULL SLAM. SESSIONS ARE PROCESSED SEQUENTIALLY FROM LEFT TO RIGHT, WITH EACH SESSION BUILDING UPON THE MAP PRODUCED BY THE PRECEDING SESSION. QUAD-EASY WAS THE PRIOR SESSION.

Policy	**Q-M**	**Q-H**	**Cloister**	**Park**
Full SLAM	0.096	0.179	0.793	0.601
Distance*	-	+0.049	<u>+0.185</u>	+0.443
Distance†	-	+0.037	<u>+0.148</u>	<u>+0.138</u>
Overlap*	-	+0.048	+0.210	<u>+0.205</u>
Overlap†	-	+0.037	+0.149	**+0.135**
Inlier Ratio*	-	+0.052	+1.999	+1.273
Inlier Ratio†	-	+0.036	<u>+0.148</u>	+0.148
Inlier RMSE*	-	+0.048	+0.317	+3.380
Inlier RMSE†	-	+0.036	+0.153	+0.367
Two Step Val.*	-0.001^{1}	<u>+0.011</u>	×	$-0.217^{1\,2}$
Two Step Val.†	-0.001^{1}	<u>+0.008</u>	×	$-0.283^{1\,2}$
Two Option*	-0.004^{1}	**+0.008**	**+0.100**	$+0.168^{1}$
Two Option†	-0.001^{1}	<u>+0.008</u>	**+0.078**	$+0.069^{1}$
Rel. Cov.*	-	+0.049	+0.204	**+0.183**
Rel. Cov.†	-	+0.035	+0.157	+0.149

Bold is best for * and †. <u>Underlined</u> is second best for * and †.
*Exact boundary was kept as defined by the policies.
†Well constrained boundary used as proposed in Section VI.
- Pure localization. No new nodes were added.
×No newly explored areas were mapped
[1]Significant portions of prior session were re-mapped.
[2]Significant portions of unexplored areas were not mapped.

localization and mapping modes, and proposed a general framework for maintaining a well-connected graph across these transitions, independent of the switching policy.

Our evaluation reveals distinct trade-offs among the policies. Geometric policies are computationally inexpensive but are best suited to simple, static environments where spatial proximity reliably indicates prior map sufficiency. Registration-based policies are efficient for registration based methods as the metrics are typically readily available. The two-step validation policy explicitly incorporates graph connectivity into the switching decision but becomes computationally prohibitive as the graph grows due to the cost of computing $\bar{\lambda}_2$. The relative covariance policy provides a balance between computational cost and accuracy, and because it relies on covariance information rather than LiDAR-specific metrics, it generalizes more readily to SLAM systems using other sensing modalities.

The policies evaluated here represent a selection of possible approaches rather than an exhaustive set, and combinations of policies may be better suited to particular environments or systems. Future work will investigate integrating the proposed framework into lifelong mapping systems, including incremental handling of dynamic objects and map updates, with the goal of enabling long-term map maintenance on resource-constrained, consumer-grade robotic platforms.

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